Subject: COMPILER DESIGN
Type: Lecture Notes
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Unit 2: Parsing Tables (LR / LALR) Notes

1. Top-Down Parsing vs Bottom-Up Parsing

Top-Down: Constructs parse tree from root to leaves (LL(1) parsers, Recursive Descent). Requires elimination of left recursion and left factoring. Bottom-Up: Shift-Reduce parsers, LR(0), SLR(1), LR(1), and LALR(1).

2. FIRST and FOLLOW Computation

FIRST(alpha): Set of terminals that begin strings derived from alpha. FOLLOW(A): Set of terminals that can appear immediately to the right of non-terminal A in some sentential form.

3. LR Parsing Table Construction

Construct canonical collection of LR items using CLOSURE and GOTO operations. Resolve Shift/Reduce and Reduce/Reduce conflicts. LALR(1) merges states with identical core items to reduce table size while preventing new shift conflicts.

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